Exact ECCp-131 representation-filter search

Outcome

The frozen Rust sweep evaluated 3,450 coordinate-filter configurations in each of six exact Certicom ECCp-131 base/size cells. No held-out cell qualified: all six recorded lower confidence ratios were below 1, and the two superficially positive small-x winners at `B=64` and `B=256` were post-addition filters that require enumerating all $\Theta(B^2)$ elliptic-curve pair sums. The exact verdict is `no_qualifying_filter_in_frozen_family`.

The mechanism tested here is a representation-aware Wagner filter: train a small coordinate-bit predicate on true four-sum relations versus a rotated independent null, freeze the winner, and evaluate it once on held-out data with a matched random-subgroup base. This is an implemented and measured lead in this repository. It is **not** established as globally novel; a claim that it has never appeared anywhere would require an independent exhaustive paper/patent/code search that this run did not perform.

Held-out results

Base	B	Selected predicate	Relation / null hits (N=8192 each)	log2 enrichment	recorded lower ratio	Enumerability
small-x	64	`pair_sum_y add`, 6 bits @ 120	130 / 118	+0.1392	0.7770	post-add, $\Theta(B^2)$
random	64	`endpoint_xor sub_either`, 7 bits @ 72	118 / 126	−0.0943	0.6587	pre-add
small-x	128	`endpoint_xor target_equal`, 7 bits @ 0	120 / 140	−0.2215	0.6093	pre-add
random	128	`endpoint_xor xor`, 6 bits @ 0	129 / 118	+0.1281	0.7705	pre-add
small-x	256	`pair_sum_x add`, 6 bits @ 56	120 / 107	+0.1647	0.7783	post-add, $\Theta(B^2)$
random	256	`endpoint_xor xor`, 6 bits @ 88	137 / 138	−0.0105	0.7130	pre-add

The SVG plots recorded values from `raw/local/result.json`; no timing result is plotted. Every one of the six A/A repeats reproduced identical counts.

End-to-end phase ledger

Phase	Exact ECCp-131 evidence	Result
Curve and subgroup	canonical curve digest `9d386a…c2db`; field 131 bits; cofactor 1	passed
Factor-base construction	small-x and distinct scalar-random bases, `B=64,128,256`	completed
Hyperparameter selection	4,096 true + 4,096 null training samples per cell; held-out labels not used	3,450 configs/cell
Held-out relation filter	8,192 true + 8,192 null samples per cell	no qualifying filter
Relation finding, arity 2	256 setup + 256 target candidates, `B=256`	0 rows, rank 0

Phase	Exact ECCp-131 evidence	Result
Relation finding, arity 3	256 setup + 256 target candidates, `B=256`	0 rows, rank 0
Relation solving	exact 16×16 modular known-answer system	rank 16; exact recovery
F4	no exact-width decomposition backend dispatched	unavailable, not measured
F5	no exact-width decomposition backend dispatched	unavailable, not measured
F6 geometric pair index	256 signed points, 32,768 pair entries/distinct points	completed
Root finding, planted	signed four-term witness `(0,+),(1,−),(2,+),(3,+)`	found and replayed
Root finding, published target	same bounded `B=128` pair index	no witness in bounded probe
Published discrete logarithm	requires sufficient natural rows and rank	not recovered

Both natural relation runs recovered all 8/8 planted controls, so their zero natural rows are bounded null observations rather than a dead-code result. The factor-base digest was:

`74dcea140d03f56c4487baa39429fe0103f9bddfd6ec949b206eed397cfcf654`

The generic rho reference derived from the exact subgroup order is about `2^65.094` group operations; it is a cost reference, not a measured clock.

Controls, determinism, and custody

- Planted signed four-sum: found and independently replayed.
- Modular solver: all 16 unknowns determined and recovered exactly.
- A/A: 6/6 selected-filter repeats were bit-identical.
- Invalid-metadata rerun comparison: timing-free projections are byte-identical; SHA-256 `5a63e30e69cd…acd817389` (full value below).
- Instrument commit: `33355ba6aed5fbb175211a757f0cc6da460f72ab`.
- Release binary SHA-256: `3a7738ab378c…4c5a7219` (full value below).
- Frozen protocol SHA-256: `6e56a761622d…e1baa4e` (full value below).
- Pre-run checkpoint bucket: `s3://crypto-autoresearcher/`.
- Checkpoint key prefix: `checkpoints/crypto/codex__eccp131-representation-filter-20261102/`.
- Checkpoint run directory: `20261102T180106Z-7ad718962f50/`.

Full hashes, in the order counted projection, release binary, frozen protocol:

```
5a63e30e69cd7414f60374f8406e665c66cbef29a210d8c96440c22acd817389
3a7738ab378c460c06775adf1829ad9a74d3d7508d0a702c4d75bb2d4c5a7219
6e56a761622da5ab41aaff666c248b0a28bb062893f319440e074e20de1baa4e
```

The admitted run earned only local `L0` isolation. Its elapsed times are descriptive and invalid for performance or speedup claims. See `DEVIATIONS.md` for backend, interval-label, and export-shape deviations. The canonical export checker retains six coverage findings rather than inventing missing executions.

Scope and next experiment

This negative result covers only simple 1–8-bit slices of five coordinate tag sources and four combiners at the frozen offsets. It does not rule out learned multibit hashes, algebraic bucket generators, larger structured factor bases, or exact summation-polynomial decomposition. The next technical step is a coordinate-visible bucket generator whose qualifying pairs can be generated in `O(B + output)` (or another proved subquadratic bound), followed by the same train/held-out/random-base controls. Exact-width F4/F5 work remains a separate backend implementation dependency.

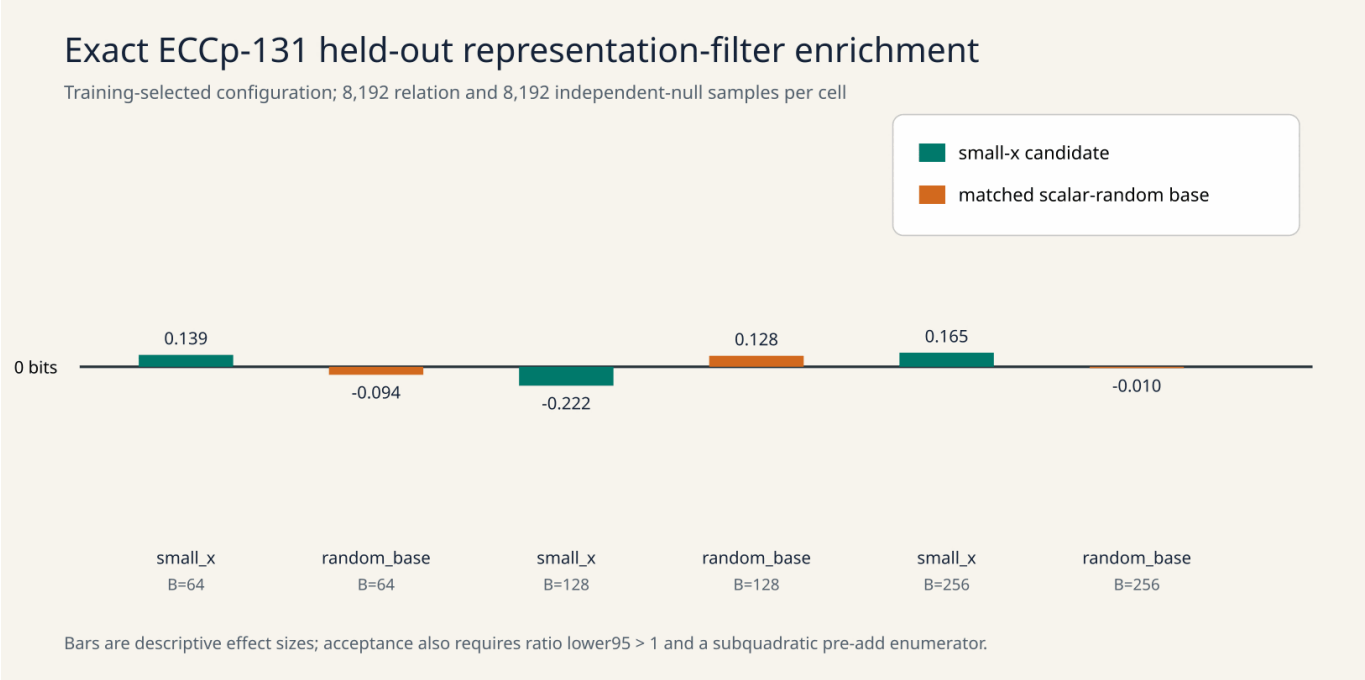


Figure 1: Held-out enrichment and recorded lower bounds

Evidence and reproduction

Raw source of truth:

- `raw/local/result.json` — sweep, A/A, root, and solver controls;
- `raw/local/ic-prime-arity2.json`, `ic-prime-arity3.json` — natural rows/rank;
- `audit/*counted-projection.json` — deterministic custody comparison;
- `experiment-export.json` — unified aggregate flow record;
- `raw/local/SHA256SUMS` and `raw/invalid-source-revision/SHA256SUMS` — inventories.

Reproduce the native sweep at the pinned source revision:

```
RAYON_NUM_THREADS=1 cargo run --release --example eccp131_representation_filter -- \
  research/eccp131_representation_filter_20261102/raw/local
```

Prior-art framing and exact-curve context are recorded in `docs/ecdlp-hardness-corpus/GALBRAITH_BOOK_AUDIT.md`, `docs/ecdlp-hardness-corpus/GENERIC_INTERFACE_AUDIT.md`, `docs/ic/CERTICOM131_IC_PROGRAM.md`, and `problems/eccp131-full-parameter-index-calculus.md`.